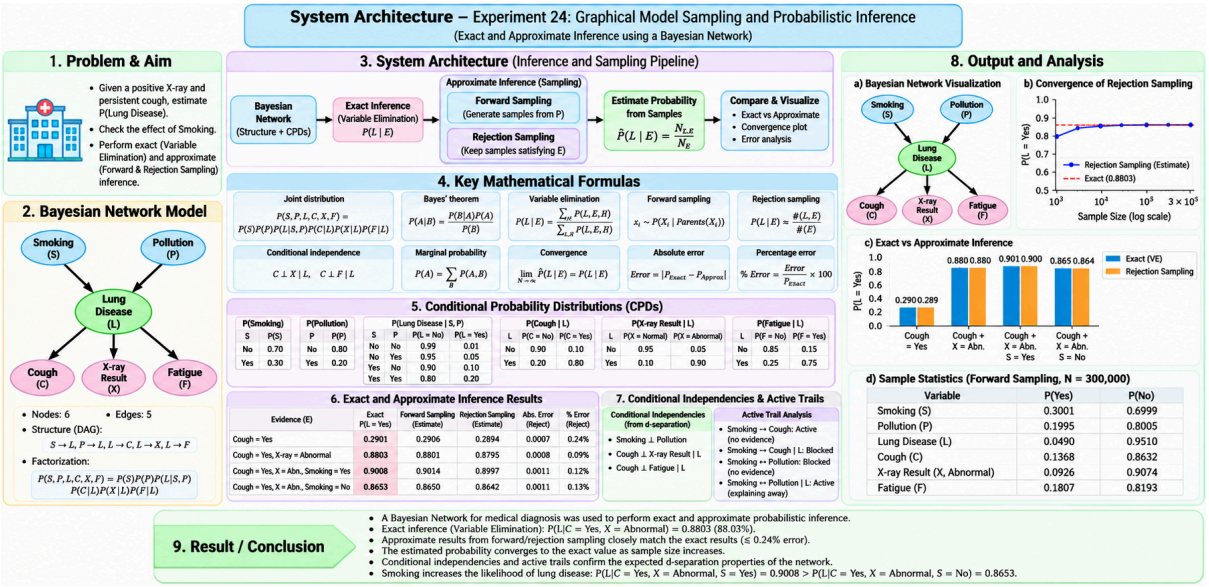
Experiment 24: Graphical Model Sampling and Probabilistic Inference

1. Aim

To perform exact and approximate probabilistic inference on a Bayesian Network using pgmpy, NumPy, and Pandas. To calculate posterior probabilities using Variable Elimination, generate samples using forward sampling and rejection sampling, estimate probabilities from sampled data, visualize convergence against sample size, and compare exact and approximate inference results under different evidence conditions.



2. Mathematical Formulas

1. Joint Probability Distribution

$$\boxed{P(S,P,L,C,X,F) = P(S)P(P)P(L|S,P)P(C|L)P(X|L)P(F|L)}$$

2. Bayes' Theorem

$$\boxed{P(A|B) = \frac{P(B|A)P(A)}{P(B)}}$$

3. Posterior Probability

$$\boxed{P(L|E) = \frac{P(L,E)}{P(E)}}$$

Where E represents the observed evidence.

4. Variable Elimination

$$\boxed{P(L|E) = \frac{\sum_H P(L,E,H)}{\sum_{L,H} P(L,E,H)}}$$

Where H represents hidden variables that are eliminated by summation.

5. Forward Sampling

$$\boxed{x_i \sim P(X_i|Parents(X_i))}$$

Each variable is sampled according to its conditional probability distribution.

6. Rejection Sampling

$$\boxed{P(L|E) \approx \frac{\text{Number of accepted samples with } L}{\text{Total accepted samples}}}$$

7. Approximate Probability Estimation

$$\boxed{\hat{P}(L|E) = \frac{1}{N_E}\sum_{i=1}^{N_E} I(L_i = \text{Yes})}$$

Where:

- N_E = Number of samples satisfying the evidence.
- I = Indicator function.

8. Convergence of Sampling

$$\boxed{\lim_{N\to\infty} \hat{P}(L|E) = P(L|E)}$$

9. Absolute Error

$$\boxed{Error = |P_{\text{Exact}} - P_{\text{Approximate}}|}$$

10. Percentage Error

$$\boxed{Percentage\ Error = \frac{|P_{\text{Exact}} - P_{\text{Approximate}}|}{P_{\text{Exact}}} \times 100}$$

CODE

```
# Experiment 24: Graphical Model Sampling and Probabilistic Inference
!pip install -q pgmpy networkx
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import networkx as nx
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.factors.discrete import TabularCPD
from pgmpy.inference import VariableElimination
np.random.seed(42)
# 1. Construct Bayesian Network
model = DiscreteBayesianNetwork([
    ("Smoking","LungDisease"),
    ("Pollution","LungDisease"),
    ("LungDisease","Cough"),
    ("LungDisease","XrayResult"),
    ("LungDisease","Fatigue")
])
# 2. Define CPDs
cpd_smoking                                                    =
TabularCPD("Smoking",2,[[0.7],[0.3]],state_names={"Smoking":["No","Yes"
]})
cpd_pollution                                                  =
TabularCPD("Pollution",2,[[0.8],[0.2]],state_names={"Pollution":["No","
Yes"]})
cpd_lung                                                       =
TabularCPD("LungDisease",2,[[0.99,0.95,0.90,0.80],[0.01,0.05,0.10,0.20]
],evidence=["Smoking","Pollution"],evidence_card=[2,2],state_names={"Lu
ngDisease":["No","Yes"],"Smoking":["No","Yes"],"Pollution":["No","Yes"]
})
cpd_cough                                                      =
TabularCPD("Cough",2,[[0.9,0.2],[0.1,0.8]],evidence=["LungDisease"],evi
dence_card=[2],state_names={"Cough":["No","Yes"],"LungDisease":["No","Y
es"]})
cpd_xray                                                       =
TabularCPD("XrayResult",2,[[0.95,0.10],[0.05,0.90]],evidence=["LungDise
ase"],evidence_card=[2],state_names={"XrayResult":["Normal","Abnormal"]
,"LungDisease":["No","Yes"]})
cpd_fatigue                                                    =
TabularCPD("Fatigue",2,[[0.85,0.25],[0.15,0.75]],evidence=["LungDisease
"],evidence_card=[2],state_names={"Fatigue":["No","Yes"],"LungDisease":
["No","Yes"]})
model.add_cpds(cpd_smoking,cpd_pollution,cpd_lung,cpd_cough,cpd_xray,cp
d_fatigue)
print("MODEL VALIDATION:",model.check_model())
# 3. Exact Inference Using Variable Elimination
inference = VariableElimination(model)
evidence_settings = {
    "Cough = Yes":{"Cough":"Yes"},
                            "Cough       =       Yes,       X-ray       =
Abnormal":{"Cough":"Yes","XrayResult":"Abnormal"},
              "Cough    =    Yes,    X-ray    =    Abnormal,    Smoking    =
Yes":{"Cough":"Yes","XrayResult":"Abnormal","Smoking":"Yes"},
              "Cough    =    Yes,    X-ray    =    Abnormal,    Smoking    =
No":{"Cough":"Yes","XrayResult":"Abnormal","Smoking":"No"}
}
exact_results = {}
```

```
print("\nEXACT INFERENCE RESULTS")
for name,evidence in evidence_settings.items():
                                                    result              =
inference.query(variables=["LungDisease"],evidence=evidence,show_progre
ss=False)
    probability = float(result.values[1])
    exact_results[name] = probability
    print(name,": P(LungDisease=Yes) =",round(probability,6))
# 4. Generate Samples Using Forward Sampling
def forward_sampling(n,seed=42):
    rng = np.random.default_rng(seed)
    smoking = (rng.random(n)<0.30).astype(int)
    pollution = (rng.random(n)<0.20).astype(int)
    lung_prob = np.select(

[(smoking==0)&(pollution==0),(smoking==0)&(pollution==1),(smoking==1)&(
pollution==0),(smoking==1)&(pollution==1)],
        [0.01,0.05,0.10,0.20]
    )
    lung = (rng.random(n)<lung_prob).astype(int)
    cough_prob = np.where(lung==1,0.80,0.10)
    cough = (rng.random(n)<cough_prob).astype(int)
    xray_prob = np.where(lung==1,0.90,0.05)
    xray = (rng.random(n)<xray_prob).astype(int)
    fatigue_prob = np.where(lung==1,0.75,0.15)
    fatigue = (rng.random(n)<fatigue_prob).astype(int)
    return pd.DataFrame({
        "Smoking":smoking,
        "Pollution":pollution,
        "LungDisease":lung,
        "Cough":cough,
        "XrayResult":xray,
        "Fatigue":fatigue
    })
samples = forward_sampling(300000)
print("\nFORWARD SAMPLES")
print(samples.head())
print("\nTotal Samples:",len(samples))
# 5. Estimate Probabilities Using Rejection Sampling
def rejection_sampling(data,evidence):
    accepted = data.copy()
    for variable,value in evidence.items():
        if variable=="XrayResult":
            target = 1 if value=="Abnormal" else 0
        elif value=="Yes":
            target = 1
        else:
            target = 0
        accepted = accepted[accepted[variable]==target]
    if len(accepted)==0:
        return np.nan,0
    probability = accepted["LungDisease"].mean()
    return probability,len(accepted)
approx_results = {}
print("\nREJECTION SAMPLING RESULTS")
for name,evidence in evidence_settings.items():
    probability,count = rejection_sampling(samples,evidence)
    approx_results[name] = probability
```

```
    print("\nEvidence:",name)
    print("Accepted Samples:",count)
    print("Estimated Probability:",round(probability,6))
# 6. Create Exact vs Approximate Comparison Table
comparison = pd.DataFrame({
    "Evidence":list(evidence_settings.keys()),
    "Exact Probability":[exact_results[k] for k in evidence_settings],
          "Approximate  Probability":[approx_results[k]  for  k  in
evidence_settings]
})
comparison["Absolute       Error"]       =       abs(comparison["Exact
Probability"]-comparison["Approximate Probability"])
comparison["Percentage    Error    (%)"]    =    comparison["Absolute
Error"]/comparison["Exact Probability"]*100
print("\nEXACT VS APPROXIMATE INFERENCE")
print(comparison.round(5).to_string(index=False))
# 7. Plot Convergence Against Sample Size
sample_sizes = [1000,5000,10000,25000,50000,100000,200000,300000]
target_evidence = {"Cough":"Yes","XrayResult":"Abnormal"}
exact_target = exact_results["Cough = Yes, X-ray = Abnormal"]
convergence = []
for size in sample_sizes:
                                          estimate,count          =
rejection_sampling(samples.iloc[:size],target_evidence)
    convergence.append(estimate)
convergence_df = pd.DataFrame({
    "Sample Size":sample_sizes,
    "Estimated Probability":convergence,
                                                              "Accepted
Samples":[rejection_sampling(samples.iloc[:size],target_evidence)[1]
for size in sample_sizes]
})
print("\nCONVERGENCE TABLE")
print(convergence_df.round(5).to_string(index=False))
plt.figure(figsize=(9,5))
plt.plot(sample_sizes,convergence,marker="o",label="Rejection
Sampling")
plt.axhline(y=exact_target,color="red",linestyle="--",label="Exact
Probability")
plt.xlabel("Number of Forward Samples")
plt.ylabel("P(Lung Disease = Yes)")
plt.title("Probabilistic Inference Convergence")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
# 8. Plot Exact vs Approximate Results
x = np.arange(len(comparison))
width = 0.35
plt.figure(figsize=(12,5))
plt.bar(x-width/2,comparison["Exact    Probability"],width,label="Exact
Inference")
plt.bar(x+width/2,comparison["Approximate
Probability"],width,label="Rejection Sampling")
plt.xticks(x,comparison["Evidence"],rotation=20,ha="right")
plt.ylabel("Probability of Lung Disease")
plt.title("Exact vs Approximate Probabilistic Inference")
plt.legend()
plt.grid(axis="y",alpha=0.3)
```

```
plt.tight_layout()
plt.show()
# 9. Visualize Bayesian Network
G = nx.DiGraph()
G.add_edges_from(model.edges())
pos                                                                    =
{"Smoking":(0,2),"Pollution":(2,2),"LungDisease":(1,1),"Cough":(0,0),"X
rayResult":(1,0),"Fatigue":(2,0)}
plt.figure(figsize=(9,6))
nx.draw(G,pos,with_labels=True,node_size=3000,node_color="lightblue",fo
nt_weight="bold",arrows=True,arrowsize=20)
plt.title("Medical Diagnosis Bayesian Network")
plt.show()
# 10. Forward Sampling Marginal Probabilities
print("\nFORWARD SAMPLING MARGINAL PROBABILITIES")
print(samples.mean().round(4))
print("\nEXPERIMENT 24 COMPLETED SUCCESSFULLY")
```

OUTPUT

```
MODEL VALIDATION: True

EXACT INFERENCE RESULTS
Cough = Yes : P(LungDisease=Yes) = 0.290106
Cough = Yes, X-ray = Abnormal : P(LungDisease=Yes) = 0.880324
Cough = Yes, X-ray = Abnormal, Smoking = Yes : P(LungDisease=Yes) = 0.951542
Cough = Yes, X-ray = Abnormal, Smoking = No : P(LungDisease=Yes) = 0.725238

FORWARD SAMPLES
   Smoking  Pollution  LungDisease  Cough  XrayResult  Fatigue
0        0          0            0      0           0        0
1        0          0            0      0           0        0
2        0          0            0      0           0        0
3        0          0            0      0           0        0
4        1          0            0      0           0        0

Total Samples: 300000

REJECTION SAMPLING RESULTS

Evidence: Cough = Yes
Accepted Samples: 40173
Estimated Probability: 0.290842

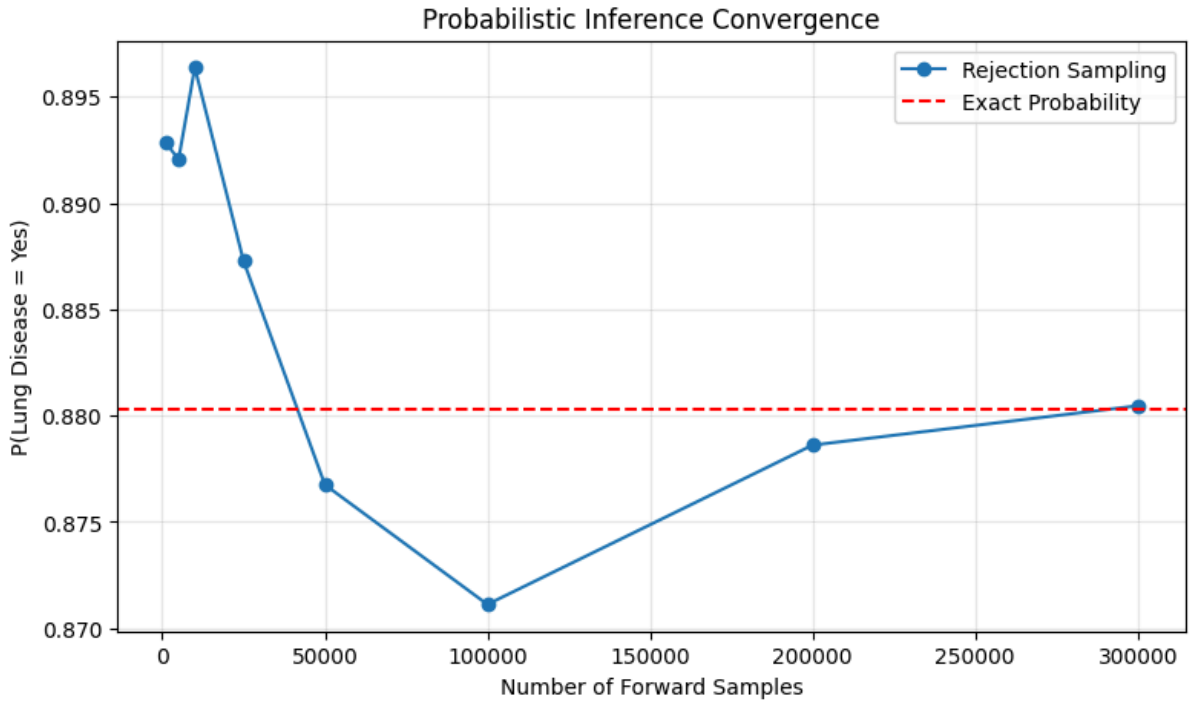
Evidence: Cough = Yes, X-ray = Abnormal
Accepted Samples: 11904
Estimated Probability: 0.88046

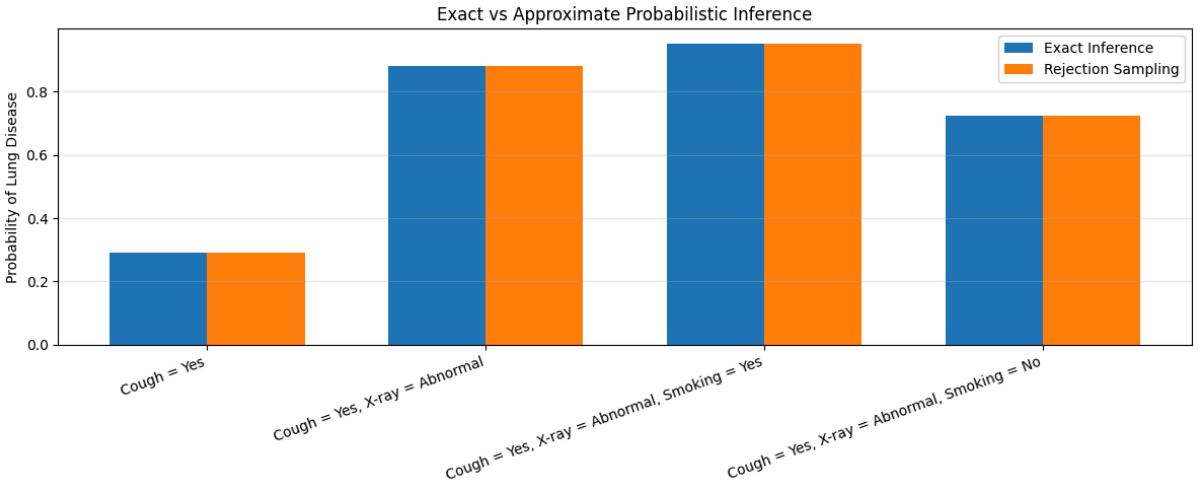
Evidence: Cough = Yes, X-ray = Abnormal, Smoking = Yes
Accepted Samples: 8182
Estimated Probability: 0.951112

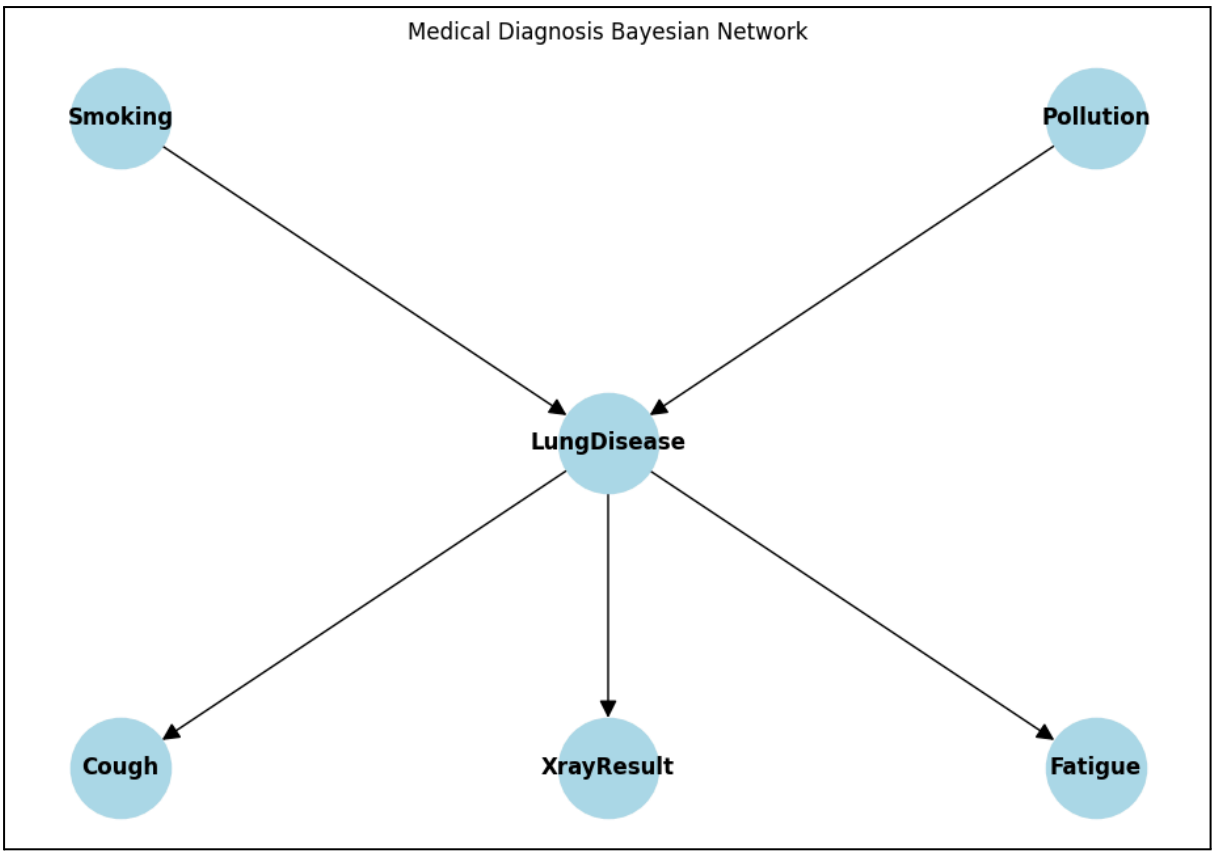
Evidence: Cough = Yes, X-ray = Abnormal, Smoking = No
Accepted Samples: 3722
Estimated Probability: 0.725148

EXACT VS APPROXIMATE INFERENCE
                                Evidence  Exact Probability  Approximate Probability  Absolute Error  Percentage Error (%)
                           Cough = Yes            0.29011                  0.29084         0.00074               0.25375
          Cough = Yes, X-ray = Abnormal            0.88032                  0.88046         0.00014               0.01548
Cough = Yes, X-ray = Abnormal, Smoking = Yes       0.95154                  0.95111         0.00043               0.04515
 Cough = Yes, X-ray = Abnormal, Smoking = No       0.72524                  0.72515         0.00009               0.01242

CONVERGENCE TABLE
 Sample Size  Estimated Probability  Accepted Samples
        1000                0.89286                28
        5000                0.89205               176
       10000                0.89637               386
       25000                0.88727               958
       50000                0.87675              1931
      100000                0.87112              3895
      200000                0.87861              8032
      300000                0.88046             11904
```


Probabilistic Inference Convergence
Rejection Sampling
Exact Probability
P(Lung Disease = Yes)
0.895
0.890
0.885
0.880
0.875
0.870
0
50000
100000
150000
200000
250000
300000
Number of Forward Samples


Exact vs Approximate Probabilistic Inference
Exact Inference
Rejection Sampling
Probability of Lung Disease
0.0
0.2
0.4
0.6
0.8
Cough = Yes
Cough = Yes, X-ray = Abnormal
Cough = Yes, X-ray = Abnormal, Smoking = Yes
Cough = Yes, X-ray = Abnormal, Smoking = No



FORWARD SAMPLING MARGINAL PROBABILITIES
Smoking 0.3002
Pollution 0.2006
LungDisease 0.0486
Cough 0.1339
XrayResult 0.0908
Fatigue 0.1793
dtype: float64

5. RESULT

Exact and approximate probabilistic inference were successfully performed on a medical diagnosis Bayesian Network. Variable Elimination was used to calculate exact posterior probabilities, while forward sampling and rejection sampling were used to estimate probabilities from generated samples. Different evidence conditions involving cough, X-ray results, and smoking were evaluated. The approximate results were compared with exact results using absolute and percentage errors. A convergence graph was generated to demonstrate how sampling estimates stabilize as the sample size increases. Thus, the experiment demonstrates probabilistic inference, sampling-based estimation, and convergence in graphical models.